Oxoacids of Phosphorous-Preparation:

Name	Formula	Oxidation state	Preparation
Hypophosphorous acid	H_3PO_2	+1	$P_4 + 6H_2O \longrightarrow 3H_3PO_2 + PH_3$
Orthophosphrous acid	H_3PO_3	+3	$P_4O_6 + 6H_2O \longrightarrow 4H_3PO_3$
Hypophosphoric acid	$H_4P_2O_6$	+4	$2P + 2O_2 + 2H_2O \longrightarrow H_4P_2O_6$
Orthophosphoric acid	H_3PO_4	+5	$P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$
Pyrophosphoric acid	$H_4P_2O_7$	+5	$2H_3PO_3 \longrightarrow H_4P_2O_7 + H_2O$

Group 16 (Oxygen group) elements:

Occurrence:

Elements belonging group 16 are called chalgogens or ore forming elements as most of the ores are oxides or sulphides. First element oxygen, the most abundant element, exists in both as dioxygen in air (above 20 % by weight as well as volume) and in combined form as oxides. Oxygen and sulphur makes up about 46.6 % & 0.034 & of earth crust by weight respectively. Sulphur exists as sulphates (gypsum, epsom etc...) and sulphide (galena, Zinc blende etc...). It is also present in the volcanic ashes. The other elements of this groups are scarce and are often found as selenides, tellurides etc... along with sulphide ores.

Physical properties:

The common physical properties of the group 16 elements are listed in the Table.

Table 3.2 Physical properties of group 16 elements

Property	Oxygen	Sulphur	Selenium	Tellurium	Polonium
Physical state at 293 K	Gas	Solid	Solid	Solid	Solid
Atomic Number	8	16	34	52	84
Isotopes	^{16}O	^{32}S	^{80}Se	^{130}Te	^{209}Po, ^{210}Po
Atomic Mass ($g.mol^{-1}$ at 293 K)	15.99	32.06	78.97	127.60	209
Electronic configuration	$[He]2s^2\ 2p^4$	$[Ne]3s^2\ 3p^4$	$[Ar]3d^{10}\ 4s^2\ 4p^4$	$[Kr]4d^{10}\ 5s^2\ 5p^4$	$[Kr]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^4$

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